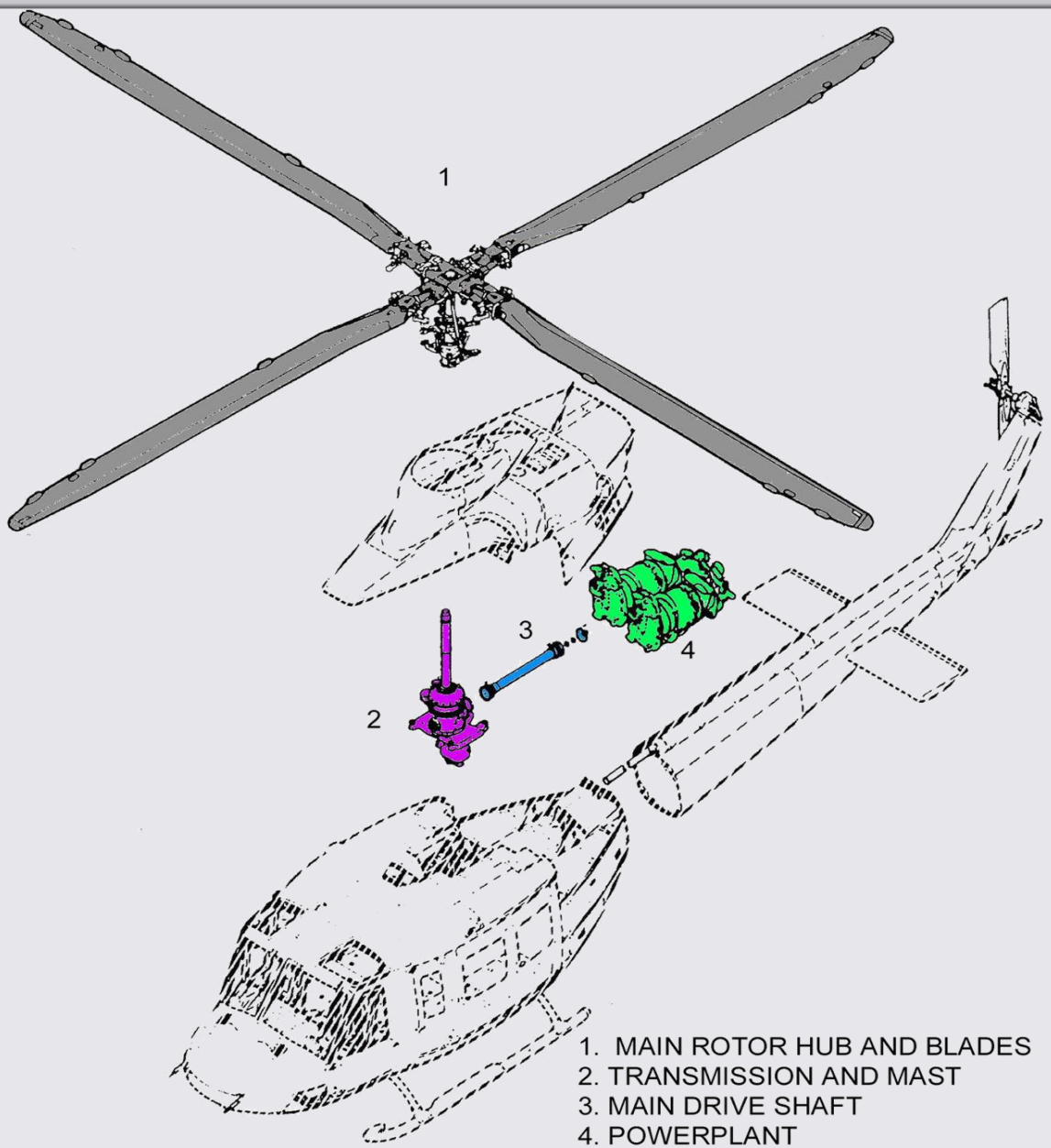
MODEL 412



1. MAIN ROTOR HUB AND BLADES
2. TRANSMISSION AND MAST
3. MAIN DRIVE SHAFT
4. POWERPLANT

INTRODUCTION

The Model 412 helicopter is a twin turbo-shaft engine helicopter, designed for high performance, low maintenance, and maximum versatility. The advanced design four-bladed rotor system permits a smoother and quieter operation along with the advantage of higher speed.

MAIN ROTOR

The main rotor is a four-bladed, soft in plane, flex-beam yoke type with blade folding capabilities. Elastomeric bearings are used to eliminate conventional lubrication requirements. The corrosion free fiberglass blades are designed for high aerodynamic efficiency and long component life.

Number of Blades	4	
Diameter	46 ft.	
Chord (Equivalent)	1 ft. 2 in.	
Disc Area	1662 sq. ft.	
Airfoil Section	8% at tip	23% at root
Tip Speed	780 ft./sec.	
RPM 100% (6600 Eng. RPM)	324 RPM	

TRANSMISSION & MAST

The transmission provides a 20.38 to 1 reduction ratio between the powerplant and the main rotor. A secondary drive train which drives the tail rotor and transmission mounted accessories will continue to operate in the event of an engine failure requiring an autorotation.

TORQUE RATING AS MEASURED AT MAIN ROTOR MAST	
Max. Continuous Power	1110 SHP
Takeoff 5 Minute Power	1370 SHP

MAIN DRIVESHAFT

The main drive shaft is connected by splined couplings between an adapter on the engine reduction gear box and the input drive quill on the transmission. Flexibility of the couplings is provided by a floating-splined method of attachment to the shaft. This method of attachment is used to accommodate movement of the transmission on pylon mounts. A spring in each coupling assists in centering of the shaft during operation and tends to extend the couplings when the shaft is removed for maintenance. Shaft couplings are attached to the engine adapter and the transmission input drive quill by bolts.

Main Driveshaft Rotational Speed (100%)	6600 RPM

POWERPLANT

The power plant is a PT6T-3D twinned turboshaft installation, consisting of two identical power sections and a common reduction gearbox. The power sections drive the output shaft to the transmission through separate reduction gear trains in the combining gearbox.

Model Number	PT6T-3D
Max. Continuous Power	1600 SHP
Takeoff Power	1800 SHP
Single Engine OEI Power	1025 SHP
Output (100%)	6600 RPM

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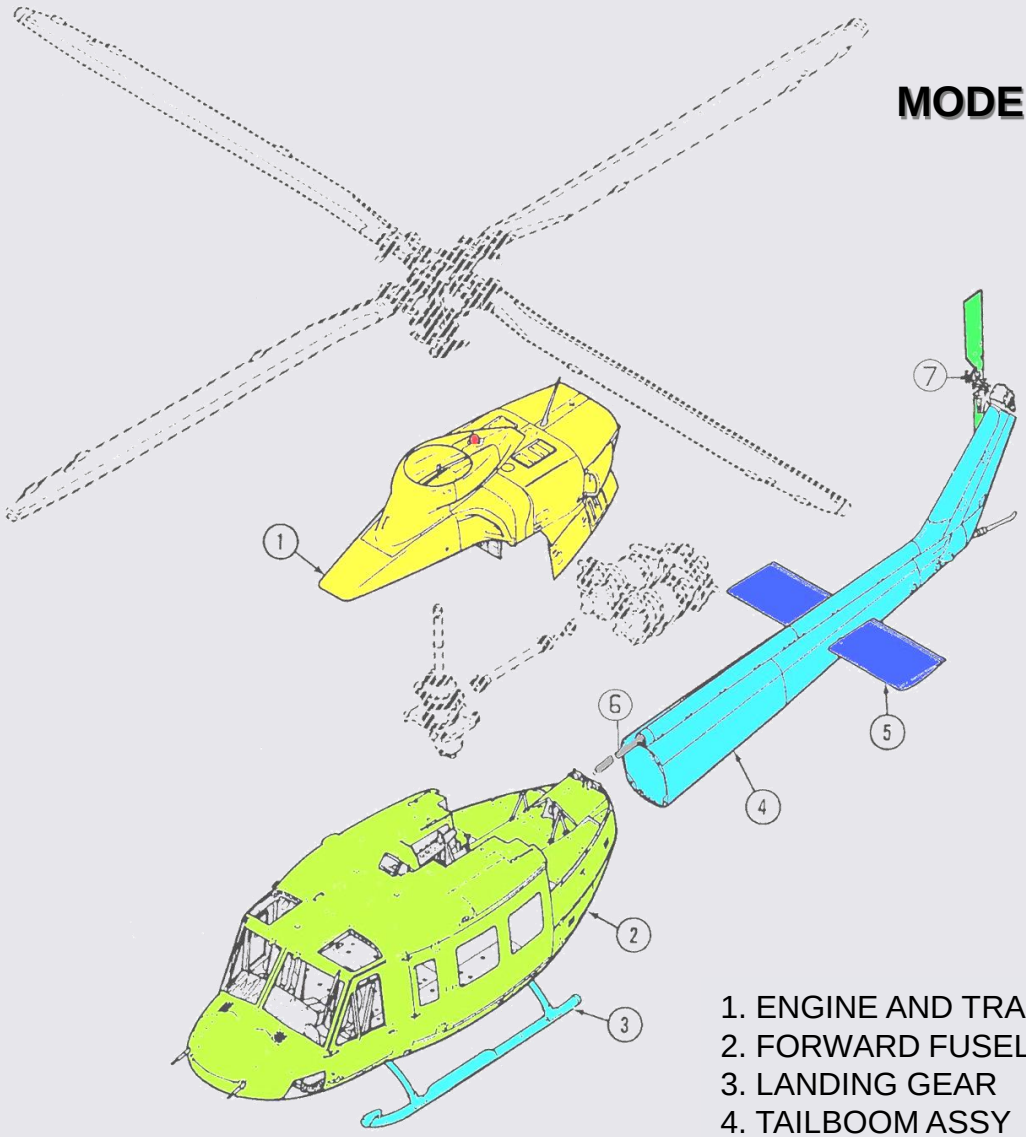


1. ENGINE AND TRANSMISSION COWLINGS
2. FORWARD FUSELAGE
3. LANDING GEAR
4. TAILBOOM ASSY
5. AERODYNAMIC ELEVATORS
6. ANTI-TORQUE DRIVE SYSTEM
7. TAILROTOR

PYLON FAIRINGS AND ENGINE COWLINGS

The pylon fairing consists of the forward transmission fairing attached to the cabin roof with quick-release fasteners and a two-piece pylon fairing mounted on the top of the forward fairing and the engine cowling.
The engine cowling consists of several sections, some secured with screws and some fitted with hinges and latches to facilitate maintenance.

FORWARD FUSELAGE

The fuselage consists of two main sections, the forward section and the aft or tailboom section. The cabin has accommodations for 15 passengers, including the pilot and co-pilot in the forward crew compartment and 13 passengers seats in the aft cabin section or passengers/cargo compartment. Conversion from passenger to cargo configuration is easily accomplished by removal of the seats. Without seats, the cabin affords approximately 220 cubic feet of cargo space.

LANDING GEAR

The landing gear is a skid-type assembly consisting of two main longitudinal tubes connected by two arched crosstubes. Each longitudinal (skid) tube is fitted with replaceable wear shoes along the bottom, a towing ring on the forward end, crew boarding step, electrically actuated passenger step, and two eyebolts for installation of the ground-handling wheels.

TAILBOOM

The tailboom assembly includes the tail rotor driveshaft covers, vertical fin, elevator, baggage compartment, and tail skid. The tailboom assembly is attached to the forward fuselage section by four high tension bolts.

ELEVATOR

The elevator consists of two separate aerodynamic surface to increase controllability and lengthen center of gravity range.

ANTI-TORQUE SYSTEM

The tail rotor drive system consists of six shafts, one tail rotor drive coupling, four bearing hanger assemblies, and two gear boxes connected in line between the transmission tail rotor drive quill and the tail rotor.

TAIL ROTOR

A single two-blade controllable pitch tail rotor is located on the right side of the tail rotor gear box. It is composed of two assemblies, the hub and the blades, and is driven by the tail rotor gear box output shaft. Blades are all metal construction and are attached by bolts to pitch change bearings in the hub yoke. The tail rotor hub is delta hinge mounted to provide for automatic equalization of thrust on advancing and retreating blades. Control links provide equal and simultaneous pitch change to both blades. The tail rotor counteracts torque of the main rotor and provides direction heading control.

Number of Blades	2
Diameter	8 ft. 7 in.
Chord	11.5 in.
Disc Area	57.8 sq. ft.
Tip Speed	745 ft./sec.
RPM 100% (6600 Eng. RPM)	1660 RPM

ELECTRICAL POWER SYSTEM

The electrical system is 28 volt D.C. single conductor system with the negative leads connected to the aircraft structure which is used as common ground. D.C. power is supplied by the generator portion of the two starter-generators operating in parallel feeding the D.C. buses. A 24 volt nickel cadmium battery can supply selected buses if both generators fail. Two 450 volt-ampere, solid state inverters supply 115/26 volts A.C. single phase, 400 hertz power to the A.C. buses.

FLIGHT INSTRUMENTS

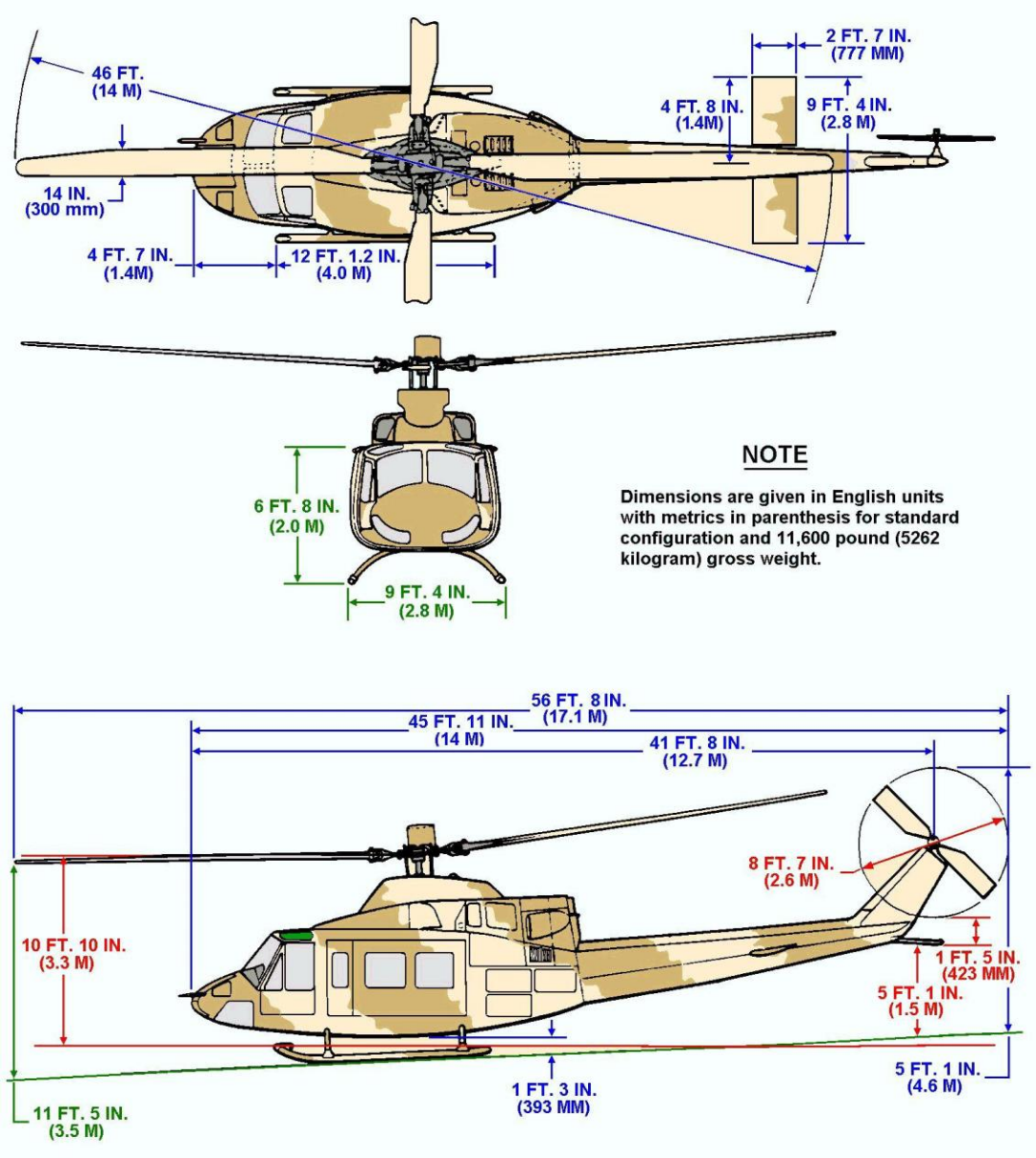
Standard type flight instruments are installed in the helicopter as basic equipment to comply with the requirements of scheduled air carrier day or night operations under Visual Flight Rule conditions.

VENTILATION

Cabin ventilation is provided by nose inlet ducts and a single inlet on the roof located under the transmission fairing.

LIGHTING

The helicopter has the required lights installed for VFR night operations, plus additional lighting equipment for utility and convenience purpose.


46 FT.
(14 M)
2 FT. 7 IN.
(777 MM)
4 FT. 8 IN.
(1.4M)
9 FT. 4 IN.
(2.8 M)
14 IN.
(300 mm)
4 FT. 7 IN.
(1.4M)
12 FT. 1.2 IN.
(4.0 M)
6 FT. 8 IN.
(2.0 M)
9 FT. 4 IN.
(2.8 M)
NOTE
Dimensions are given in English units with metrics in parenthesis for standard configuration and 11,600 pound (5262 kilogram) gross weight.
56 FT. 8 IN.
(17.1 M)
45 FT. 11 IN.
(14 M)
41 FT. 8 IN.
(12.7 M)
8 FT. 7 IN.
(2.6 M)
10 FT. 10 IN.
(3.3 M)
1 FT. 5 IN.
(423 MM)
5 FT. 1 IN.
(1.5 M)
5 FT. 1 IN.
(4.6 M)
1 FT. 3 IN.
(393 MM)
11 FT. 5 IN.
(3.5 M)

412HP S/N 33214-33999 & 36020-36086 DIFFERENCES

ENGINE

Model Number	PT6T-3BE
Manufacturer	Pratt & Whitney of Canada Limited
Max. Continuous Power	1600 SHP
Takeoff Power	1800 SHP
Single Engine OEI Power	1025 SHP
Output (100%)	6600 RPM

412SP S/N 33108-33213 & 36001-36019 DIFFERENCES

ENGINE

Model Number	PT6T-3B
Manufacturer	Pratt & Whitney of Canada Limited
Max. Continuous Power	1600 SHP
Takeoff Power	1800 SHP
Single Engine OEI Power	1025 SHP
Output (100%)	6600 RPM

TORQUE RATING AS APPLIED TO THE TRANSMISSION

Max. Continuous Power	1134 SHP
Takeoff 5 Minute Power	1400 SHP

412 S/N 33107 & PRIOR DIFFERENCES

TORQUE RATING AS APPLIED TO THE TRANSMISSION

Max. Continuous Power	1134 SHP
Takeoff 5 Minute Power	1350 SHP

WEIGHTS

Standard Configuration (Approx. Empty Weight)	6200 lbs
Maximum Gross Weight	11,600 lbs

FUEL

Capacity (Total)	220 U.S. Gals.
(Usable)	214 U.S. Gals.

412 REVIEW
INTRODUCTION

1. The maximum gross weight is limited to ___11,900___ pounds on the skids and ___11,900___ pounds in flight with an external load.

2. The main rotor is described as a soft-in-plane, ___FLEX -BEAM___.

3. The main rotor diameter is ___46___ ft.

4. The 412 consists of two major sections, the forward ___FUSELAGE___ and ___TAILBOOM___.

5. The 412 can carry two crew members and ___13___ passengers.